

Multikernel Linux's new release provides kernel isolation without hypervisors

Multikernel Linux has released mklinux v7.0-mk2, its first public test release, as an open source project aimed at providing an alternative to conventional virtual machines (VMs) and containers. It allows multiple independent Linux kernels to run simultaneously on a single bare-metal machine without a hypervisor or container service.

A host kernel divides pools of central processing units (CPUs), memory, and peripheral component interconnect (PCI) devices into instances, with each



instance booting and running its own kernel natively on allocated resources. There is no emulation or hypervisor layer intercepting operations, while only explicitly provided resources are shared.

Instances are defined through device trees, with planned overlays allowing

CPUs, memory, and devices to move dynamically between the resource pool and running instances without restarting. Resources can also be reclaimed when an instance is shut down.

The project is designed to provide stronger isolation than containers while avoiding some VM overhead. It can potentially assign separate kernels to applications and physical-device drivers, with virtual network and block devices enabling communication.

Initial measurements from Cong Wang, founder and CEO, Multikernel Project, show performance gains of 1.07× to 2.5× over kernel-based virtual machine (KVM) guests, depending on the workload. Context switching was 2.5× faster, while memory bandwidth and latency were reportedly similar. “Compared to VMs, there is no second layer of page tables and no device model,” Wang said.

The project targets artificial intelligence (AI) training and inference, latency-sensitive services, and workloads requiring strong isolation. It is initially available for x86_64, based on Linux Kernel 7.0, through its public GitHub repository.

Linux Foundation takes AIRSEAI into its fold

The LF AI & Data Foundation, an umbrella organisation under the Linux Foundation, has welcomed AIRSEAI, an open source embodied artificial intelligence project initiated by the Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS). The move provides AIRSEAI with a neutral environment for global collaboration on deployable embodied AI, where artificial intelligence systems interact with the physical world.

AIRSEAI aims to address fragmentation in the robotics ecosystem, where proprietary platforms, hardware-specific software stacks, isolated datasets and models can make systems difficult to evaluate, reuse and deploy across different machines. Under the LF AI & Data Foundation, AIRSEAI will provide a unified framework intended to allow researchers, developers and manufacturers to test and deploy AI across diverse hardware architectures without vendor lock-in.

The project follows a three-stage roadmap covering modular single-robot intelligence, data-driven dual-arm manipulation and collective adaptive intelligence. AIRSEAI 2.0 adds production-oriented capabilities such as vision-based tactile sensing. Future AIRSEAI 3.0 updates are planned to enable different types of robots to coordinate across distributed environments.

OpenClaw 2.0 revamps personal AI agent setup

OpenClaw has released version 2026.8.1, described as OpenClaw 2.0, with major changes to the setup and operation of its open source personal artificial intelligence agent. The update introduces a more guided onboarding process, expands the role of the browser interface and adds shared cloud sessions that allow authorised users to continue work through the same Gateway.

The new setup process can detect supported AI access already available on a user's machine, including existing ChatGPT or Claude command-line sign-ins, API keys, configured model routes and compatible local models running through Ollama or LM Studio. OpenClaw tests the selected route with an actual completion request before saving the configuration, helping users confirm that their chosen model connection works correctly.

OpenClaw 2.0 also makes its browser application a central workspace for conversations, settings and ongoing agent activities. Users can return to existing work and monitor agent actions through the browser, while the Gateway continues to hold the main conversation state, credentials, permissions and other durable information.

A significant technical change involves session storage. Existing file-based sessions and transcripts can be migrated to SQLite during the update process. The release also introduces shared cloud sessions, allowing authorised collaborators to join or take over work stored by a Gateway. Remote workers can execute tasks using the shared session state without directly receiving provider credentials. OpenClaw's role and permission system controls access, although the project makes clear that users sharing a Gateway remain within the same trust domain rather than operating in fully isolated environments.